BIF604 – Special Topics in Bioinformatics

Final Term – 2026 (Solved)

MCQs

1. Polony technology is … **polymerase colony**
2. Binary alignment/map files (BAM) are the preferred **SRA** submission format.
3. The progressive approach **multiple alignment** of a group of sequences first align ost similar pair.
4. Network Analyzer allows for the creation of connected components. **sub-networks**
5. There is. difference between modules and read packages? **NO**
6. Python concatenates all lines together into a single line, and end-of-line **\n** character is added where line break.
7. The Gene Association files provide an annotation only for the **Gene Ontology** Z test is... **n>30**
8. Which of the following is the categories of gene ontology? **biological processes, molecular functions, cellular component**
9. The data is displayed in a grid where each row represents a ... and each column represents a...? **Gene, sample**
10. Need to transform the fluorescence intensity image obtained after scanning into a measurement of gene expression for each spot.? **Microarray Image processing**
11. Comprises a set of methods to remove systematic experimental errors and bias? **Microarray-Normalization**
12. Gene Expression analyzing gene expression involves; **Detection of mRNA present in a cell or tissue at a particular point in time**
13. Network Analyzer allows for the creation of .... of connected components. **Sub-networks**
14. Cytoscape is _ software? 1) freeware 2) online **3) modular software** 4) all
15. In Python, ... is used to introduce escape sequences in the code**. Backslash **
16. When our Python scripts have crashed we ‘ve assumed there is something wrong with the script and tried to fix it. **Exception**
17. Affymetrix ..... nucleotides in length. **25**
18. A ...is a programming language construct for creating a list based on existing
19. lists. **list comprehension**
20. are written as series of items in parenthesis. **Tuple**
21. RE uses backslash characters …… to indicate special meanings. **(\)**
22. Gene Ontology (GO) term enrichment is a technique for interpreting sets of genes making use of the Gene Ontology system of classification, in which genes are assigned to a set of predefined bins depending on them? **functional characteristic**

23. Gene Ontology (GO) term enrichment is a technique for interpreting sets of genes according to their - **Functional characteristic**
24. Network Analyzer allows for the creation of ..... of connected components - **sub-networks**
25. Python class is used to implement .....defined data types. **user**
26. Study of evolutionary relationships between - **Genes**
27. Affymetrix nucleotides in length - **25**
28. Types of loop - **2 types**
29. Cytoscape has the ability to associate a Nested Network/ with any node -**Cluster**
30. ___ comprises a set of methods to remove systematic experimental errors and bias - **Normalization microarray**
31. A _ incorporates an independent network that does not share nodes or attributes with the current network - **Nested Network**
32. Gene association file Provide annotation- **only gene ontology** -
33. pythons are language -**modular**
34. Unweighted Pair – Group Method using arithmetic Averages) is the abbreviation of - **UPGMA**
35. Probe size and spacing determines the _ of the array**? resolution**
36. comprises asset of methods to remove systematic experimental errors and bias - **Microarray Normalization**
37. Strings are enclosed **in either single ('') or double ("")** quotation marks.
38. Need to transform the fluorescence intensity image obtained after scanning into a measurement of gene expression for each spot - **Microarray Image processing**
39. T test is___ **n<30**
40. Unrooted collection of unique elements___? **Set**
41. Python concatenates all lines together into a single line, and end-of-line \n character is added where line break.
42. A …… is a set of genes (nodes) connected by edges representing functional relationships among these genes. **Gene network**
43. A component which tends to enhance or suppress the working of correspondent genes is known.... **Regulator**

Short Questions:

1. Define Heat map? Write its two functions.

A common method of visualizing gene expression data is to display it as a heat map. A **heat map** is a graphical representation of data in which values are shown as different **colors**.
In bioinformatics/microarray analysis, it is commonly used to display **gene expression levels**, where:

- Warm colors (e.g., red) → high expression
- Cool colors (e.g., blue/green) → low expression

Functions:

1. **Visualization of data patterns**: Converts complex numerical data into color-coded form, makes it easy to see trends, clusters, or differences between samples/genes
2. **Comparison and identification**: Helps compare expression levels across conditions (normal vs diseased), Identifies up-regulated or down-regulated genes quickly

2. Why we create list in python language?

Lists are created in Python to:

- Store multiple values,
- organize related data,
- Easy access & modification,
- Supports iteration (loops) and
- Flexible data types.

3. Role of Phylogenetic tree in MSA?

- Displays evolutionary relationships inferred from aligned sequences.
- Helps identify conserved and variable regions.
- Guides interpretation of functional and structural similarities.

4. Describe types of microarray.

- cDNA microarray.
- Oligonucleotide microarray (e.g., Affymetrix).
- Protein microarray.

5. DAVID Annotation Tool

Database for Annotation, Visualization and Integrated Discovery (DAVID) is a Free Bioinformatics Resource Software. It Aims to provide functional interpretation of large datasets of genes derived from different genomic studies such as microarray.

This software Classifies large gene lists into functionally related groups so Genes are highly organized so that they are readable and understandable. It also ranks the importance of the functional gene groups which helps a quick focus on only the significant ones.

6. Objective programing

Object-Oriented Programming (OOP) is a programming paradigm in which programs are designed using **objects and classes** instead of only functions and procedures.
An **object** represents a real-world entity that contains **data (attributes)** and **functions (methods)**.

It helps organize code into reusable and manageable structures.

Key Features (Principles)

Encapsulation — Bundling data and methods together
Inheritance — Creating new classes from existing ones
Polymorphism — Same function behaving differently
Abstraction — Showing essential features, hiding details

7. what is BINGO

A *Biological Network Gene Ontology tool* (BiNGO) is a Java-based tool to determine which Gene Ontology (GO) categories are statistically overrepresented in a set of genes or a subgraph of a biological network.

BiNGO is implemented as a plugin for Cytoscape, which is an open source bioinformatics software platform for visualizing and integrating molecular interaction networks.

It helps interpret biological meaning from network data.

8. One-line code for open file in python

- open ("filename.txt", "r") — opens file in read mode.
- Returns a file object for reading/writing.
- Mode "w" for writing, "a" for append.

9. Bio python

BioPython

The Biopython Project is an international association of developers of freely available Python (http://www.python.org) tools for computational molecular biology. The Biopython web site (http://www.biopython.org) provides an online resource for modules, scripts, and web links for developers of Python-based software for bioinformatics use and research.

Python is easy to learn, has a very clear syntax and can easily be extended with modules written in C, C++ or FORTRAN

10. Gene network

A gene interaction network is a set of genes (nodes) connected by edges representing functional relationships among these genes.

The basic components of any network are

• **Nodes**; Nodes represent units in the network,

• **Edges;** edges represent the interactions between the units

Visualizing all of the attributes—localization (color), expression level (size) and expression correlation (edge thickness), helps generate consensus results about the gene networks under observation.

11. R- language

- R is a programming language and software environment for statistical analysis, graphics representation and reporting. R was created by Ross Ihaka and Robert Gentleman at the University of Auckland, New Zealand, and is currently developed by the R Development Core Team.
- R, a programming language for statistical data analysis, the plots and tables produced by R can readily be used in typewriting systems such as Emacs, LATEX, or Word.
- R is freely available under the GNU General Public License, and pre-compiled binary versions are provided for various operating systems like Linux, Windows and Mac.

12. How the interface of Cytoscape look like? Explain also feature.

Interface of Cytoscape

Interface:
The Cytoscape interface is a **graphical user interface (GUI)** designed to visualize and analyze biological networks. It typically contains:

- **Menu bar & Toolbar** — for opening files, layout control, and analysis tools
- **Network View Panel** — main area showing nodes and edges of the network
- **Control Panel (left)** — manage networks, styles, filters
- **Table Panel (bottom)** — displays node/edge data attributes

Features

- **Network visualization:** Displays molecular interaction networks graphically
- **Data integration:** Combines gene/protein data with networks
- **Analysis tools:** Supports clustering, pathway analysis
- **Modular plugins/apps:** Extend functionality
- **Custom layouts & styling:** Change node color, size, shape

13. Define Filtered List Comprehensions.

Filtered List Comprehensions

Definition:
A **filtered list comprehension** is a concise way to create a **new list** in Python by including only those elements from an existing list that satisfy a certain **condition**.

Syntax

new_list = [expression for item in iterable if condition]

- expression → value to include in the new list
- item → element from the iterable
- condition → filter that determines which items are included

Example

```
numbers = [1, 2, 3, 4, 5, 6]
even_numbers = [x for x in numbers if x % 2 == 0]
print(even_numbers)
```

Output:

[2, 4, 6]

Only numbers divisible by 2 are included in the new list.

14. What is microarray normalization

Microarray normalization

• Process to remove systematic variations in microarray data.

• Ensures accurate comparison of gene expression across arrays.

• Steps: data cleaning → within-array normalization → between-array normalization.

15. Emerging role of Python in bioinformatics

- Automation of biological data analysis workflows.
- Integration with bioinformatics libraries like Biopython.
- Supports statistical, visualization, and sequence analysis tasks.

Long Questions:

1. **Four ways to create network in Cytoscape?**

Creating Networks

There are 4 different ways of creating networks in Cytoscape:

• **Importing pre-existing, fixed-format network files.** Networks are imported into Cytoscape through the File → Import → Network menu. (e.g., XGMML, SIF).

• **Importing pre-existing, unformatted text or Excel files.** Cytoscape supports the import of networks from delimited text files and Excel workbooks using File → Import → Network → File....

• **Importing data from from public databases.** Users can access various kinds of databases through this function, under File → Import → Network → Public Databases.... (e.g., STRING, BioGRID).

• **Creating an empty network and manually adding nodes and edges.** To create an empty network, go to File → New → Network → Empty Network, and then manually add network components by right clicking on the network canvas or on a node.

2. **How modules are imported.**

- Use import module name to load.
- from module import function to import specific functions.
- Supports built-in, third-party, or custom modules.

3. **Describe the array Applications.**

- Used for high-throughput gene expression profiling.
- Detects mRNA levels for thousands of genes simultaneously.
- Supports SNP genotyping and mutation detection.
- Applied in disease classification and drug target discovery.
- Enables comparative genomic hybridization for copy number variation.
- Facilitates biomarker identification in clinical research.

4. **Microarray application.**

Applications of Microarrays

Gene expression analysis

- Measure expression levels of thousands of genes at once

- Compare normal vs diseased tissues

Disease diagnosis & prognosis

- Identify disease-related genes
- Used in cancer classification and prediction

Drug discovery & toxicology

- Study gene response to drugs
- Identify drug targets and side effects

Genotyping / SNP detection

- Detect genetic variations
- Used in population genetics and personalized medicine

Comparative genomics

- Compare gene sequences between species
- Study evolution and gene function

Pathogen detection

- Identify bacteria or viruses
- Used in clinical and environmental monitoring

Mutation & copy number analysis

- Detect gene deletions or duplications
- Important in genetic disorder studies

5. Loops of python

Loops are used in Python to execute a block of code repeatedly until a condition is met. They help automate repetitive tasks and reduce code length.

Python mainly has **two types of loops**:

1. for Loop

for loop provides an easy way to access elements of the string.

Example

```
>>> DNA = "ATGCAGCT"
```

>>> for base in DNA: print (base, end="")

print ("\noutside loop")

Output: ATGCAGCT

Common Uses:

- Traversing lists/strings
- Counting iterations
- Performing repeated operations

2. while Loop

Repeats execution **as long as a condition is true**.

Example

```
i = 1
while i <= 5:
    print(i)
    i += 1
```

Important Note

- If condition never becomes false → **infinite loop**

6. One-line code of python for simple input and output commands

One-line code for Input

name = input("Enter your name: ")

One-line code for Output

print(name)

7. Define

- **Bonferroni correction:** Pre-analysis stage to assess raw data relationships. The **Bonferroni correction** is a statistical method used to reduce **false positive results (Type I errors)** when multiple hypothesis tests are performed simultaneously.
- **FDR:** (False Discovery Rate): expected proportion of false positives in results.
  • Controls Type I error in multiple hypothesis testing.

- **Qfdr:** adjusted p-value controlling FDR.
  - Lower q-values indicate more reliable results.
  - Used in high-throughput studies like microarrays.

8. What are the type of line that indicate in the network?

Types of Lines (Edges) in a Network

In a network, **lines (also called edges)** represent interactions or relationships between nodes (genes, proteins, or molecules). The type of line indicates the **nature of the interaction**. Common types include:

1. **Directed line (arrow)**

- Shows **direction of interaction**
- Example: Gene A activates Gene B

2. **Undirected line (simple line)**

- Shows **interaction without direction**
- Example: Protein–protein binding

3. **Dashed line**

- Often represents **indirect or predicted interactions**

4. **Colored line**

- Different colors can indicate **activation (green), inhibition (red), or other functional types**

9. How many operations are there for list in Python?

Operations on Lists in Python

1. **Accessing Elements** – Using index

```
mylist[0]   # first element
```

2. **Slicing** – Extract part of a list

```
mylist[1:4]
```

3. **Modifying Elements** – Change values by index

mylist[2] = 10

4. **Adding Elements**

- append() → add single element
- extend() → add multiple elements
- insert() → add at specific index

5. **Removing Elements**

- remove() → remove by value
- pop() → remove by index
- clear() → remove all elements

6. **Searching Elements**

- index() → find position
- count() → count occurrences

7. **Sorting / Reversing**

- sort() → sort list
- reverse() → reverse order

8. **Copying**

- copy() → shallow copy

9. **Other Operations**

- Concatenation: +
- Repetition: *
- Membership test: in

10. Evolutionary tree from a matrix in MSA

- Start with a multiple sequence alignment (MSA).
- Calculate distance matrix from sequence similarities.
- Apply algorithms like Neighbor-Joining or UPGMA.
- Build a tree showing branching relationships.
- Root tree if evolutionary direction is known.
- Visualize using phylogenetic software.

11. Write a note on Gene networks and its components.

A gene interaction network is a set of genes (nodes) connected by edges representing functional relationships among these genes.

The basic components of any network are

- **Nodes**; Nodes represent units in the network,
- **Edges;** edges represent the interactions between the units
- **Regulatory Elements**; Transcription factors, promoters, enhancers, Control when and how genes are expressed
- **Signals/Inputs**; Environmental or cellular signals, Trigger regulatory responses within the network
- **Output Responses;** Final gene expression products (RNA/proteins), Lead to cellular functions or phenotypic changes

Visualizing all of the attributes—localization (color), expression level (size) and expression correlation (edge thickness), helps generate consensus results about the gene networks under observation.

12. How do you make a set in Python?

- Use set () constructor or curly braces {} to create.
- Automatically removes duplicates.
- Supports union (|), intersection (&), and difference (-).
- Elements must be immutable (e.g., numbers, strings).
- Used for fast membership testing.
- Dynamic and unordered data collection.

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